

Grade 6 Accelerated
Unit 13
Lesson 1 Practice

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1. What is the value of the expression $6 \cdot 2 - (x - 4)$ if $x = 1$?

$$\begin{aligned} &6 \cdot 2 - (1 - 4) \\ &12 - 1 + 4 \\ &11 + 4 \\ &15 \end{aligned}$$

2. Find the value of $8 - 2(4 + y) + 2y \cdot 2$ when $y = 3$.

$$\begin{aligned} &8 - 2(4 + 3) + 2(3) \cdot 2 \\ &8 - 8 - 6 + 6 \cdot 2 \\ &\cancel{8} - \cancel{8} - 6 + 12 \\ &6 \end{aligned}$$

3. Evaluate $2p + \frac{16}{j} - 11$ for $p = 5$ and $j = 8$.

$$\begin{aligned} &2(5) + \frac{16}{8} - 11 \\ &10 + 2 - 11 \\ &12 - 11 \\ &1 \end{aligned}$$

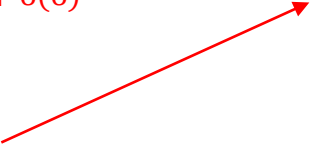
4. Evaluate $10 - 20 \div 4f + 11$ if $f = 2$.

$$\begin{aligned} &10 - 20 \div 4(2) + 11 \\ &10 - 5(2) + 11 \\ &\cancel{10} - \cancel{10} + 11 \\ &11 \end{aligned}$$

5. Find the value of $7 \cdot (2m - 4m + 18)$ when $m = 3$.

$$\begin{aligned} &7(2 \cdot 3 - 4 \cdot 3 + 18) \\ &7(6 - 12 + 18) \\ &7(-6 + 18) \\ &7(12) \\ &84 \end{aligned}$$

6. Find the value of $7 - (y + 12)^2 + 6m$ if $m = 6$ and $y = 2$.

$$\begin{aligned} &= 7 - (2 + 12)^2 + 6(6) &= -189 + 36 \\ &= 7 - (14)^2 + 36 &= -153 \\ &= 7 - 196 + 36 \end{aligned}$$


7. Find the value of $12g - (16 - u - 8) \div 2$ if $g = 3$ and $u = 11$.

$$\begin{aligned} &12(3) - (16 - 11 - 8) \div 2 \\ &12(3) + (-16 + 11 + 8) \div 2 \\ &12(3) + (-5 + 8) \div 2 \\ &12(3) + (3) \div 2 \\ &36 + 3 \div 2 \\ &36 + 1.5 \\ &37.5 \end{aligned}$$

8. Evaluate $8a + 4b(b + 10)$ for $a = 3$ and $b = 2$.

$$\begin{aligned} &8(3) + 4(2)(2 + 10) \\ &8(3) + 4(2)(12) \\ &24 + 8(12) \\ &24 + 96 \\ &120 \end{aligned}$$

9. What is the value of the expression $5g \cdot h^3 + (h - 1)^2$ if $g = 1$ and $h = 2$?

$$\begin{aligned} &= 5(1) \cdot 2^3 + (2 - 1)^2 \\ &= 5 \cdot 8 + (1)^2 \\ &= 40 + 1 \\ &= 41 \end{aligned}$$

10. What is the value of the expression $\left(\frac{b^2-6}{4}\right) \cdot 7 + y^2$ if $b = 2$ and $y = 5$?

$$\begin{aligned} &= \left(\frac{2^2-6}{4}\right) \cdot 7 + 5^2 &= -\frac{1}{2} \cdot \frac{7}{1} + 25 \\ &= \left(\frac{4-6}{4}\right) \cdot 7 + 25 &= -3.5 + 25 \\ &= \left(\frac{-2}{4}\right) \cdot 7 + 25 &= 21.5 \end{aligned}$$


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Lesson 2 Practice

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1. Evaluate the expression $8b + 10 - (9 - 6)^2$ for $b = -5$.

$$\begin{aligned} &= \underbrace{8(-5)} + 10 - \underbrace{(9 - 6)}^2 \\ &= -40 + 10 - \underbrace{(3)}^2 \\ &= -40 + 10 - 9 \\ &= -30 - 9 = -39 \end{aligned}$$

2. Evaluate $-7m - 5q^2 + \frac{s}{2}$ when $m = -8$, $q = -2$, and $s = 56$.

$$\begin{aligned} &= \underbrace{-7(-8)} - 5\underbrace{(-2)^2} + \frac{56}{2} \\ &= 56 - 5(4) + 28 \\ &= 56 - 20 + 28 \\ &= 36 + 28 = 64 \end{aligned}$$

3. Evaluate $2s - 6q - f^2$ for $s = -8$, $q = 10$, and $f = -12$.

$$\begin{aligned} &= \underbrace{2(-8)} - \underbrace{6(10)} - \underbrace{(-12)^2} \\ &= -16 - 60 - (144) \\ &= -76 - 144 \\ &= -220 \end{aligned}$$

4. Evaluate $-m^3 - 3d + (n - y)$ if $m = 4$, $d = -6$, $n = -9$, and $y = 0$.

$$\begin{aligned} &= \underbrace{-(4)^3} - \underbrace{3(-6)} + \underbrace{(-9 - 0)} \\ &= -\underbrace{(64)} + 18 + (-9) \\ &= -46 - 9 \\ &= -55 \end{aligned}$$

5. Evaluate $4g^4 - \frac{z}{4}$ when $g = -4$ and $z = 24$.

$$\begin{aligned} &= 4(-4)^4 - \frac{(24)}{4} \\ &= 4(256) - \frac{24}{4} \\ &= 1024 - 6 \\ &= 1018 \end{aligned}$$

6. What is the value of $5(8x + 4) - 3m^2$ if $x = 3$ and $m = 15$?

$$\begin{aligned} &= 5(8(3) + 4) - 3(15)^2 \\ &= 5(24 + 4) - 3(225) \\ &= 5(28) - 3(225) \\ &= 140 - 675 \\ &= -535 \end{aligned}$$

7. Find the value of $-9(h - 6) + 2a - (9 - a)^5$ if $h = 6$, and $a = 12$.

$$\begin{aligned} &= -9(6 - 6) + 2(12) - (9 - 12)^5 \\ &= \cancel{-9(0)} + 24 - (-3)^5 \\ &= 24 - (-243) \\ &= 267 \end{aligned}$$

8. Evaluate the expression $p^3 - 5k(y - k)$ for $p = 6$, $k = -4$ and $y = 1$.

$$\begin{aligned} &= 6^3 - \underbrace{5(-4)}_{(-20)} \underbrace{(1 - (-4))}_{(5)} \\ &= 216 - (-20)(5) \\ &= 216 - (-100) \\ &= 316 \end{aligned}$$

9. Evaluate the expression $8 \div j - 12 \div (-10 + 4)$ if $j = -4$.

$$\begin{aligned} &= \underbrace{8 \div (-4)}_{-2} - \underbrace{12 \div (-10 + 4)}_{12 \div (-6)} \\ &= -2 - (-2) \\ &= 0 \end{aligned}$$

10. What is the value of the expression $4m - 10 + (a - 11) \cdot 4^3$ if $m = 5$ and $a = -1$?

$$\begin{aligned} &= 4(5) - 10 + \underbrace{(-1 - 11)}_{(-12)} \cdot 4^3 \\ &= 20 - 10 + (-12) \cdot 64 \\ &= \underbrace{20 - 10}_{10} + (-768) \\ &= 10 + (-768) \\ &= -758 \end{aligned}$$

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Lesson 3 Practice

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Determine if each pair of expressions are equivalent.

1. $-2(10 + 4)$ and $-2(10) + 2(4)$
 $-2(14)$ $-20 + 8$
 -28 $\neq$ -12
 not equivalent

2. $0.5(5 - 2)$ and $0.5 \cdot 5 + 0.5 \cdot -2$
 $= 0.5(3)$ $= 2.5 - 1$
 1.5 $=$ 1.5
 equivalent

3. $\frac{3}{4} \times \frac{9}{10}$ and $\frac{9}{10} \times \frac{3}{4}$
 $= \frac{3}{4} \times \frac{9}{10}$ $= \frac{9}{10} \times \frac{3}{4}$
 $= \frac{27}{40}$ $= \frac{27}{40}$
 equivalent

4. $\left(\frac{2}{3} + \frac{1}{8}\right) + \frac{1}{6}$ and $\frac{2}{3} + \left(\frac{1}{8} + \frac{1}{6}\right)$
 $8\left(\frac{2}{3}\right) + \left(\frac{1}{8}\right) 3 + \left(\frac{1}{6}\right) 4$ $8\left(\frac{2}{3}\right) + \left(\frac{1}{8}\right) 3 + \left(\frac{1}{6}\right) 4$
 $\left(\frac{16}{24} + \frac{3}{24}\right) + \frac{4}{24}$ $\frac{16}{24} + \left(\frac{3}{24} + \frac{4}{24}\right)$
 $\frac{19}{24} + \frac{4}{24} = \frac{23}{24}$ $\frac{16}{24} + \left(\frac{7}{24}\right) = \frac{23}{24}$
 equivalent

5. $\frac{1}{6} + \frac{2}{9}$ and $\frac{2}{9} + \frac{1}{6}$
 $\frac{3}{3}\left(\frac{1}{6}\right) + \left(\frac{2}{9}\right) \frac{2}{2}$ $\frac{2}{2}\left(\frac{2}{9}\right) + \left(\frac{1}{6}\right) \frac{3}{3}$
 $\frac{3}{18} + \frac{4}{18}$ $\frac{4}{18} + \frac{3}{18}$
 $\frac{7}{18}$ $=$ $\frac{7}{18}$
 equivalent

6. $\frac{7}{11} - \frac{1}{3}$ and $\frac{1}{3} - \frac{7}{11}$
 $\frac{3}{3}\left(\frac{7}{11}\right) - \left(\frac{1}{3}\right) \frac{11}{11}$ $\frac{11}{11}\left(\frac{1}{3}\right) - \left(\frac{7}{11}\right) \frac{3}{3}$
 $\frac{21}{33} - \frac{11}{33}$ $\frac{11}{33} - \frac{21}{33}$
 $\frac{10}{33} \neq \frac{-10}{33}$
 not equivalent

7. $\frac{5}{7} + \left(\frac{1}{2} + \frac{3}{4}\right)$ and $\left(\frac{5}{7} + \frac{1}{2}\right) + \frac{3}{4}$
 $4\left(\frac{5}{7}\right) + \frac{14}{14}\left(\frac{1}{2}\right) + \left(\frac{3}{4}\right) \frac{7}{7}$ $\left(\frac{20}{28} + \frac{14}{28}\right) + \frac{21}{28} = \frac{55}{28}$
 $\frac{20}{28} + \left(\frac{14}{28} + \frac{21}{28}\right) = \frac{20}{28} + \frac{35}{28} = \frac{55}{28}$
 equivalent

8. $\left(\frac{1}{5} - \frac{2}{5}\right) - \frac{8}{9}$ and $\frac{1}{5} - \left(\frac{2}{5} - \frac{8}{9}\right)$
 $\left(\frac{1}{5} - \frac{2}{5}\right) - \frac{8}{9}$ $\frac{1}{5} - \left(\frac{2}{5} - \frac{8}{9}\right)$
 $-\frac{1}{5} - \frac{8}{9}$ $\frac{1}{5} - \left(\frac{18}{45} - \frac{40}{45}\right)$
 $-\frac{9}{45} - \frac{40}{45} = \frac{-49}{45}$ $\frac{1}{5} - \left(-\frac{22}{45}\right)$
 $\frac{9}{45} + \frac{22}{45} = \frac{31}{45}$
 not equivalent

9. Rewrite $(6x - 2y) + 9$ using the associative property.
 $6x + (-2y + 9)$

10. Rewrite $9z - 7b$ using the commutative property.
 $-7b + 9z$

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Lesson 4 Practice

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1. Rewrite the expression $(-5h + 8) + 2h$ using the associative property.
 $-5h + (8 + 2h)$
2. Rewrite the expression $(-5h + 8) + 2h$ using the commutative property.
Answers can vary.
 $2h + (-5h + 8)$
3. Rewrite the expression $8 \cdot (-7 \cdot 3y)$ using the associative property.
 $(8 \cdot -7)3y$
4. Rewrite the expression $8 \cdot (-7 \cdot 3y)$ using the commutative property.
Answers can vary.
 $(-7 \cdot 3y) \times 8$
5. Rewrite the expression $-2r + 4(11k + 3)$ using the distributive property.
Answers can vary.
 $-2r + (4 \times 11k) + (4 \times 3)$
6. Rewrite the expression $-2r + 4(11k + 3)$ using the commutative property of multiplication.
Answers can vary.
 $-r2 + 4(11k + 3)$
7. Rewrite the expression $-2r + 4(11k + 3)$ using the commutative property of addition.
Answers can vary.
 $-2r + 4(3 + 11k)$

Write two equivalent algebraic expressions for each expression.

- | | | | | | |
|----|--------------------|----|----------------|-----|------------------|
| 8. | $(2 + 4) \cdot 7$ | 9. | $7t \cdot 2$ | 10. | $-6x + 6 - 18$ |
| | $7(2) + 7(4)$ | | $2 \cdot 7t$ | | $(-6x + 6) - 18$ |
| | $(4 + 2) \times 7$ | | $7(t \cdot 2)$ | | $-6x + (6 - 18)$ |

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Lesson 5 Practice

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Write an algebraic expression for each written description.

1. The quotient of a number and nine.

$$n \div 9 \text{ or } \frac{n}{9}$$

2. The product of a number and eleven.

$$n \times 11 \text{ or } 11n$$

3. Three-fifths more than a number.

$$n + \frac{3}{5} \text{ or } \frac{3}{5} + n$$

4. A number doubled.

$$n \times 2 \text{ or } 2n$$

5. The difference of a twenty and a number.

$$20 - n$$

6. The school provides each student, s , 5 hall passes per quarter.

$$5 \times s \text{ or } 5 \cdot s \text{ or } 5s$$

7. There are 8 snack bags, each with b , number of snacks in each bag.

$$8 \times b \text{ or } 8 \cdot b \text{ or } 8b$$

8. A hair salon makes 18 per haircut, h .

$$18 \times h \text{ or } 18 \cdot h \text{ or } 18h$$

9. An electricity company charges \$0.11 per kilowatt, k .

$$0.11 \times k \text{ or } 0.11 \cdot k \text{ or } 0.11k$$

10. A football team scored 14 points less than the other team, t .

$$t - 14$$

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Lesson 6 Practice

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Write an algebraic expression to represent each real-world situation.

1. Julia has 11 videogames downloaded to her device. She deleted x of those games from the device. How many videogames are left on her device?

$$11 - x$$

2. Tony has x tickets from a local arcade. His brother has 10 times more tickets than Tony. How many tickets does his brother have?

$$10x \text{ or } 10 \cdot x$$

3. Susan sold 42 candy bars for a fundraiser. Her friend sold x more candy bars than her. How many candy bars did the friend sell?

$$42 + x \text{ or } x + 42$$

4. A grocery store has received a shipment of 60 new boxes of cereal to be placed equally on x shelves. How many boxes will be on each shelf?

$$60 \div x \text{ or } \frac{60}{x}$$

5. Harold has a plant that measured h inches tall last week. This week, the plant is 19 inches tall.

$$19 - h$$

Answer may vary.

For each expression, create a real-world scenario that the algebraic expression could represent.

6. $x + 7$

Andrea earned \$7 more than her younger brother earned, x .

7. $\frac{22}{x}$

A bag of 22 pieces of candy is split evenly among x friends.

8. $71y$

71 grams of dry food are given to each of the small dogs, y , at the shelter.

9. $k - 81$

Cherie's weekly budget, k , is decreased by \$81 after grocery shopping.

10. $12 + h$

Jack has h homework problems to complete. His friend from another class has 12 more problems than Jack has to do.

For questions 1-2, draw a representation of the expression with algebra tiles. Cross out any zero pairs. Then write an equivalent algebraic expression after eliminating any zero pairs.

	Algebraic Expression and Representation	Equivalent Algebraic Expression
1.	$-8x - 2 + 6x$ Representation:	$-2x - 2$
2.	$10 + z - 6 + z$ Representation:	$2z + 4$

Write an equivalent expression by combining like terms for questions 3-10.

3. $4x + 9x$

$13x$

4. $-2y + 5 + y - 7$

$-2y + y - 7 + 5$

$-y - 2$

5. $r + -r + r - r$

$r + r - r - r$

0

6. $6y - 3 + 8 - 4y$

$6y - 4y + 8 - 3$

$2y + 5$

7. $y - 4 + 5y + 8$

$y + 5y + 8 - 4$

$6y + 4$

8. $10x + 9 + x - 1$

$10x + x + 9 - 1$

$11x + 8$

9. $-5y + 2 + 3 + 9y$

$9y - 5y + 2 + 3$

$4y + 5$

10. $8 + z - 10 + 2z$

$z + 2z - 10 + 8$

$3z - 2$

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Lesson 8 Practice

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Find the sum of each expression using the fewest terms possible.

1. $(0.5x + 0.1) + (0.6x - 0.9)$

$$(0.5x + 0.6x) + (0.1 - 0.9)$$

$$1.1x - 0.8$$

2. $\left(-\frac{2}{5} + 7y\right) + \left(4y - \frac{3}{5}\right)$

$$(7y + 4y) + \left(-\frac{2}{5} - \frac{3}{5}\right)$$

$$11y - 1$$

3. $(-0.3x) + (x + 9)$

$$(x - 0.3x) + 9$$

$$0.7x + 9$$

4. $\left(5\frac{1}{2}y - \frac{4}{5}\right) + \left(8\frac{1}{5} - 3\frac{3}{4}y\right)$

$$\left(5\frac{1}{2}y - 3\frac{3}{4}\right) + \left(8\frac{1}{5} - \frac{4}{5}\right)$$

$$4\frac{6}{4}$$

$$7\frac{6}{5}$$

$$1\frac{3}{4}y + 7\frac{2}{5}$$

5. $\left(10 - \frac{2}{3}y\right) + \left(-7 + \frac{2}{3}y\right)$

$$\left(\frac{2}{3}y - \frac{2}{3}y\right) + (10 - 7)$$

$$3$$

Determine the value of each expression.

6. $(2.7 + 8.4x) + (-3.6x + 7.5) =$

$$(8.4x - 3.6x) + (2.7 + 7.5)$$

$$4.8x + 10.2$$

7. $\left(\frac{1}{8}x + \frac{3}{4}\right) + \left(\frac{3}{4}x - \frac{7}{16}\right) =$

$$\left(\frac{3}{4}x + \frac{1}{8}x\right) + \left(\frac{3}{4} - \frac{7}{16}\right)$$

$$\left(\frac{6}{8}x + \frac{1}{8}x\right) + \left(\frac{12}{16} - \frac{7}{16}\right)$$

$$\frac{7}{8}x + \frac{5}{16}$$

8. $(-2.9y + 6) + (5.9 + 4y) =$

$$(4y - 2.9y) + (6 + 5.9)$$

$$1.1y + 11.9$$

9. $\left(1.0 + \frac{4}{9}x\right) + (-6.6 + x) =$

$$\left(\frac{4}{9}x + x\right) + (1 - 6.6)$$

$$1\frac{4}{9}x - 5.6$$

10. $\left(11.8 + \frac{3}{7}x\right) + \left(-\frac{11}{7} + x\right) =$

$$\left(\frac{3}{7}x + x\right) + \left(11.8 - \frac{11}{7}\right)$$

$$\left(11\frac{56}{70} - 1\frac{40}{70}\right)$$

$$10\frac{16}{70} \rightarrow 10\frac{8}{35}$$

$$1\frac{3}{7}x + 10\frac{8}{35}$$

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Lesson 9 Practice

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Find the difference of each expression using the fewest terms possible.

1. $(0.6x + 0.7) - (0.8x - 0.9)$

$$(0.6x + 0.7) - 0.8x + 0.9$$

$$(0.6x - 0.8x) + (0.7 + 0.9)$$

$$-0.2x + 1.6$$

2. $(-\frac{4}{5} + 9y) - (3y - \frac{3}{5})$

$$(-\frac{4}{5} + 9y) - 3y + \frac{3}{5}$$

$$9y - 3y + (\frac{3}{5} - \frac{4}{5})$$

$$6y + (-\frac{1}{5})$$

$$6y - \frac{1}{5}$$

3. $(-0.2z) - (z + 8)$

$$-0.2z - z - 8$$

$$-1.2z - 8$$

4. $(8\frac{1}{2}y - \frac{2}{5}) - (6\frac{3}{5} - 2\frac{3}{4}y)$

$$(8\frac{1}{2}y - \frac{2}{5}) - 6\frac{3}{5} + 2\frac{3}{4}y$$

$$(8\frac{1}{2}y + 2\frac{3}{4}y) + (-\frac{2}{5} - 6\frac{3}{5})$$

$$11\frac{1}{4}y + (-7)$$

$$11\frac{1}{4}y - 7$$

5. $(10 - \frac{4}{9}q) - (-8 - \frac{2}{9}q)$

$$(10 + (-\frac{4}{9}q)) + 8 + \frac{2}{9}q$$

$$(\frac{2}{9}q - \frac{4}{9}q) + (10 + 8)$$

$$-\frac{2}{9}q + 18$$

Simplify each of the expressions.

6. $(7.6 - 4.9x) - (2.8x - 3.7) - (9.8x - 4.2)$

$$7.6 - 4.9x - 2.8x + 3.7 - 9.8x + 4.2$$

$$-4.9x - 2.8x - 9.8x + 7.6 + 3.7 + 4.2$$

$$-17.5x + 15.5$$

7. $\left(\frac{1}{8}x - \frac{3}{4}\right) + \left(\frac{3}{4}x + \frac{13}{16}\right) + \left(\frac{3}{8}x - \frac{1}{4}\right)$

$$\frac{1}{8}x - \frac{12}{16} + \frac{6}{8}x + \frac{13}{16} + \frac{3}{8}x - \frac{4}{16}$$

$$\frac{1}{8}x + \frac{6}{8}x + \frac{3}{8}x - \frac{12}{16} + \frac{13}{16} - \frac{4}{16}$$

$$\frac{10}{8}x - \frac{3}{16}$$

$$\frac{5}{4}x - \frac{3}{16}$$

$$1\frac{1}{4}x - \frac{3}{16}$$

8. $(2.8y + 8) - (2.7 + 3y) + (-8.7 + 5y)$

$$2.8y + 8 - 2.7 - 3y - 8.7 + 5y$$

$$2.8y - 3y + 5y + 8 - 2.7 - 8.7$$

$$4.8y - 3.4$$

9. $\left(2.8 + \frac{4}{9}r\right) + (-6.8 + r) - \left(-4.7 + \frac{7}{9}r\right)$

$$2.8 + \frac{4}{9}r - 6.8 + r + 4.7 - \frac{7}{9}r$$

$$\frac{4}{9}r + r - \frac{7}{9}r + 2.8 - 6.8 + 4.7$$

$$\frac{6}{9}r + 0.7$$

$$\frac{2}{3}r + 0.7$$

10. $\left(\frac{5}{6} + \frac{3}{7}n\right) - \left(-\frac{1}{6} + n\right) - \left(2\frac{5}{6} + 7\frac{2}{7}n\right)$

$$\frac{5}{6} + \frac{3}{7}n + \frac{1}{6} - n - 2\frac{5}{6} - 7\frac{2}{7}n$$

$$\frac{3}{7}n - n - 7\frac{2}{7}n + \frac{5}{6} + \frac{1}{6} - 2\frac{5}{6}$$

$$\frac{3}{7} - \frac{7}{7} - \frac{51}{7} = -\frac{55}{7} = -7\frac{6}{7}$$

$$\frac{5}{6} + \frac{1}{6} - \frac{17}{6} = -\frac{11}{6} = -1\frac{5}{6}$$

$$-7\frac{6}{7}n - 1$$

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Determine if the expressions are equivalent for the given values.

	Value	$8x + 4x - 5$	$2(6x - 8) + 11$	Equivalent?
1.	$x = -4$	$8(-4) + 4(-4) - 5$ $-32 + (-16) - 5$ $-48 - 5$ -53	$2(6(-4) - 8) + 11$ $2(-24 - 8) + 11$ $-48 + (-16) + 11$ $-64 + 11$ -53	☒ yes ☐ no
2.	$x = 0$	$8(0) + 4(0) - 5$ $0 + 0 - 5$ -5	$2(6 \cdot 0 - 8) + 11$ $2(0 - 8) + 11$ $2(-8) + 11$ $-16 + 11$ -5	☒ yes ☐ no
3.	$x = 0.5$	$8(0.5) + 4(0.5) - 5$ $4 + 2 - 5$ 1	$2(6 \cdot 0.5 - 8) + 11$ $2(3 - 8) + 11$ $2(-5) + 11$ $-10 + 11$ 1	☒ yes ☐ no
4.	$x = \frac{1}{3}$	$8\left(\frac{1}{3}\right) + 4\left(\frac{1}{3}\right) - 5$ $\frac{8}{3} + \frac{4}{3} - 5$ $\frac{12}{3} - 5$ $4 - 5$ -1	$2\left(6 \cdot \frac{1}{3} - 8\right) + 11$ $2\left(\frac{6}{3} - 8\right) + 11$ $2(2 - 8) + 11$ $2(-6) + 11$ $-12 + 11$ -1	☒ yes ☐ no

5. Depending on the value being substituted into the expressions, the evaluated

expressions

☒ always
☐ sometimes
☐ never

had the same value as each other.

Determine whether the expressions are equivalent for the given values.

		$\frac{1}{5}(15 + 20y) + 2x$	$4\left(2x + y + \frac{3}{4}\right)$	Equivalent?
6.	$x = -2$ $y = 6$	$\frac{1}{5}(15 + 20 \cdot 6) + 2(-2)$ $\frac{1}{5}(15 + 120) - 4$ $\frac{1}{5}(135) - 4$ $27 - 4 = 23$	$4\left(2(-2) + 6 + \frac{3}{4}\right)$ $4\left(-4 + 6 + \frac{3}{4}\right)$ $4(2.75)$ 11	☐ yes ☒ no
7.	$x = 0$ $y = 1$	$\frac{1}{5}(15 + 20 \cdot 1) + 2 \cdot 0$ $\frac{1}{5}(15 + 20) + 0$ $\frac{1}{5}(35) + 0$ 7	$4\left(2 \cdot 0 + 1 + \frac{3}{4}\right)$ $4\left(0 + 1 + \frac{3}{4}\right)$ $4(1.75)$ 7	☒ yes ☐ no
8.	Answers will vary $x = 2$ $y = 0$	$\frac{1}{5}(15 + 20 \cdot 0) + 2(2)$ $\frac{1}{5}(15 + 0) + 4$ $\frac{1}{5}(15) + 4$ $3 + 4 = 7$	$4\left(2 \cdot 2 + 0 + \frac{3}{4}\right)$ $4\left(4 + 0 + \frac{3}{4}\right)$ $4(4.75)$ 19	☐ yes ☒ no

Determine if the following algebraic expressions are equivalent using any method you choose.

	Expression A	Expression B	Equivalent?
9.	$x(13 - 15) - \frac{3}{2}(4 + 6x)$ $-2x - 6 - 9x$ $-11x - 6$ $-11(0) - 6$ -6	$-11x - 15$ $-11(0) - 15$ -15	☐ yes ☒ no
10.	$\frac{5}{7}(2x + 12x) - (6^2)$ $\frac{5}{7}(14x) - 36$ $10x - 36$ $10(0) - 36$ -36	$-36 + 10x$ $-36 + 10(0)$ $-36 + 0$ -36	☐ yes ☒ no

Grade 6 Accelerated
Unit 13
Lesson 11 Practice

Name Answer Key
Date _____
Period _____

From the specified set of integers, determine which, if any, make the equation true.

$\{-10, 1, 3, 12\}$

1. $20 - 5w = 10w + 5$
 $20 - 5(-10) = 10(-10) + 5$
 $20 + 50 = -100 + 5$
 $70 \neq -95$ NOT TRUE

$20 - 5(1) = 10(1) + 5$
 $20 - 5 = 10 + 5$
 $15 = 15$ TRUE

$20 - 5(3) = 10(3) + 5$
 $20 - 15 = 30 + 5$
 $5 \neq 35$ NOT TRUE

$20 - 5(12) = 10(12) + 5$
 $20 - 60 = 120 + 5$
 $-40 \neq 125$ NOT TRUE

2. $4 + x = 2x - 8$
 $4 + (-10) = 2(-10) - 8$
 $-6 = -20 - 8$
 $-6 \neq -28$ NOT TRUE

$4 + (1) = 2(1) - 8$
 $5 = 2 - 8$
 $5 \neq -6$ NOT TRUE

$4 + (3) = 2(3) - 8$
 $7 = 6 - 8$
 $7 \neq -2$ NOT TRUE

$4 + (12) = 2(12) - 8$
 $16 = 24 - 8$
 $16 = 16$ TRUE

3. $3y + 5 = 4y + 15$
 $3(-10) + 5 = 4(-10) + 15$
 $-30 + 5 = -40 + 15$
 $-25 = -25$ TRUE

$3(1) + 5 = 4(1) + 15$
 $3 + 5 = 4 + 15$
 $8 \neq 19$ NOT TRUE

$3(3) + 5 = 4(3) + 15$
 $9 + 5 = 12 + 15$
 $14 \neq 27$ NOT TRUE

$3(12) + 5 = 4(12) + 15$
 $36 + 5 = 48 + 15$
 $41 \neq 63$ NOT TRUE

4. $2p - 6 = 6p - 18$
 $2(-10) - 6 = 6(-10) - 18$
 $-20 - 6 = -60 - 18$
 $-26 \neq -78$ NOT TRUE

$2(1) - 6 = 6(1) - 18$
 $2 - 6 = 6 - 18$
 $-4 \neq -12$ NOT TRUE

$2(3) - 6 = 6(3) - 18$
 $6 - 6 = 18 - 18$
 $0 = 0$ TRUE

$2(12) - 6 = 6(12) - 18$
 $24 - 6 = 72 - 18$
 $18 \neq 54$ NOT TRUE

Determine if the inequality $18 < 2m - 6$ is true for each value.

5. 8
 $18 < 2(8) - 6$
 $18 < 16 - 6$
 $18 < 10$
NOT TRUE

6. 12
 $18 < 2(12) - 6$
 $18 < 24 - 6$
 $18 < 18$
NOT TRUE

7. 15
 $18 < 2(15) - 6$
 $18 < 30 - 6$
 $18 < 24$
TRUE

Determine if the inequality $-y + 3 + 8y > -3y - 7$ is true for each value.

8. -5
 $-(-5) + 3 + 8(-5) > -3(-5) - 7$
 $5 + 3 - 40 > 15 - 7$
 $-32 > 8$
NOT TRUE

9. 0
 $-(0) + 3 + 8(0) > -3(0) - 7$
 $0 + 3 + 0 > 0 - 7$
 $3 > -7$
TRUE

10. 5
 $-(5) + 3 + 8(5) > -3(5) - 7$
 $-5 + 3 + 40 > -15 - 7$
 $38 > -22$
TRUE

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Lesson 12 Practice

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Write an equivalent expression using the associative, commutative, or distributive property. State the property used. **answers may vary**

1. $6x - \frac{2}{7}(21x - 35)$

$6x - 6 + 10$

distributive property

2. $(11x + 15) - 3x + 7$

$(15 + 11x) - 3x + 7$

commutative property

3. $\left(-\frac{3}{5}y + 2\right) \cdot 20$

$20 \cdot \left(-\frac{3}{5}y + 2\right)$

commutative property

4. $-24\left(\frac{3}{4}x + 1\right) + (18 - 3x)$

$-18x - 24 + (18 - 3x)$

distributive property

5. If expressions are not equivalent, then they will

- ☐ always
- ☐ sometimes
- ☒ never

have the same terms when they are simplified using properties of operations.

Use properties of operations to determine if the expressions are equivalent.

6. $0.5x - 0.1(5x - 2)$ and $7x - (-2 + 7x)$

$$0.5x - 0.5x + 0.2 = 0.2$$

$$7x + 2 - 7x = 2$$

not equivalent

7. $(4x + 12) - 6$ and $7x - 3(x - 2)$

$$4x + 12 - 6 = 4x + 6$$

$$7x - 3x + 6 = 4x + 6$$

equivalent

8. $2\frac{1}{5}x - 8 + x$ and $\frac{16}{5}\left(x - \frac{5}{2}\right)$

$$2\frac{1}{5}x + x - 8 = \frac{16}{5}x - 8$$

$$\frac{16}{5}x - \frac{16}{2} = \frac{16}{5}x - 8$$

equivalent

9. $0.3 + 0.1(17x - 4)$ and $2x - \left(\frac{3}{5}x - \frac{2}{5}\right)$

$$0.3x + 1.7x - 0.4 = 1.7x - 0.1$$

$$2x - \frac{3}{5}x + \frac{2}{5} = 1.4x + 0.4$$

not equivalent

10. Expressions that are equal are not equivalent.

☐ True

☒ False